



Joshnavi K B 251801122-2025-AIDS ▾

J2

Started on Thursday, 6 August 2026, 10:08 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:20 AM**Time taken** 11 mins 58 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int c=0;
6      int i=1;
7      c++;
8      int s=1;
9      c++;
10 while(1){
11     c++;
12     if(s<=n){
13         i++;
14         c++;
15         s+=i;
16         c++;
17     }
18     else{
19         break;
20     }
21 }
22 printf("%d",c);
24 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Thursday, 6 August 2026, 10:39 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:47 AM**Time taken** 7 mins 31 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6
7      scanf("%d", &n);
8
9      if (n == 1)
10 {
11     printf("1");
12 }
13 else
14 {
15     int count = 5 * n + 2;
16     printf("%d", count);
17 }
18
19     return 0;
20 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓

	Input	Expected	Got	
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on Thursday, 6 August 2026, 10:49 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:58 AM**Time taken** 9 mins 26 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int c=0;
6      c++;
7      for(int i=1;i<=n;++i){
8          c++;
9          if(n%i==0){
10             c++; //printf("%d",i);
11         }
12         c++;
13     }
14     printf("%d",c);
15 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on Thursday, 6 August 2026, 11:04 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:09 AM**Time taken** 4 mins 49 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int c=0;
6      c++;
7      for(int i=n/2;i<n;i++){
8          c++;
9          for(int j=1;j<n;j=2*j){
10             c++;
11             for(int k=1;k<n;k*=2){
12                 c++;
13                 c++;
14             }
15             c++;
16         }
17         c++;
18     }
19     c++;
20     printf("%d",c);
21 }
22 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on Thursday, 6 August 2026, 11:11 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:17 AM**Time taken** 6 mins 26 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int c=0;
6      int rev=0;
7      c++;
8      int rem;
9      c++;
10 while(n!=0){
11     c++;
12     rem=n%10;
13     c++;
14     rev=rev*10+rem;
15     c++;
16     n/=10;
17     c++;
18 }
19 c++;
20 printf("%d",c);
21 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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